

for all unit vectors x . This result follows from C. Borell's lemma, which itself is a consequence of Brunn-Minkowski inequality; see [81, Section 2.2.b₃].

Exercise 3.4.10. ☹☹ Show that the concentration inequality in Theorem 3.1.1 may not hold for a general isotropic sub-gaussian random vector X . Thus, independence of the coordinates of X is an essential requirement in that result.

3.5 Application: Grothendieck's inequality and semidefinite programming

In this and the next section, we use high-dimensional Gaussian distributions to pursue some problems that have seemingly nothing to do with probability. Here we give a probabilistic proof of Grothendieck's inequality, a remarkable result which we will use later in the analysis of some computationally hard problems.

Theorem 3.5.1 (Grothendieck's inequality). *Consider an $m \times n$ matrix (a_{ij}) of real numbers. Assume that, for any numbers $x_i, y_j \in \{-1, 1\}$, we have*

$$\left| \sum_{i,j} a_{ij} x_i y_j \right| \leq 1.$$

Then, for any Hilbert space H and any vectors $u_i, v_j \in H$ satisfying $\|u_i\| = \|v_j\| = 1$, we have

$$\left| \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \right| \leq K,$$

where $K \leq 1.783$ is an absolute constant.

There is apparently nothing random in the statement of this theorem, but our proof of this result will be probabilistic. We will actually give two proofs of Grothendieck's inequality. The one given in this section will yield a much worse bound on the constant K , namely $K \leq 288$. In Section 3.7, we present an alternative argument that yields the bound $K \leq 1.783$ as stated in Theorem 3.5.1.

Before we pass to the argument, let us make one simple observation.

Exercise 3.5.2. ☹

- (a) Check that the assumption of Grothendieck's inequality can be equivalently stated as follows:

$$\left| \sum_{i,j} a_{ij} x_i y_j \right| \leq \max_i |x_i| \cdot \max_j |y_j|. \quad (3.12)$$

for any real numbers x_i and y_j .

- (b) Show that the conclusion of Grothendieck's inequality can be equivalently stated as follows:

$$\left| \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \right| \leq K \max_i \|u_i\| \cdot \max_j \|v_j\| \quad (3.13)$$

for any Hilbert space H and any vectors $u_i, v_j \in H$.

Proof of Theorem 3.5.1 with $K \leq 288$. **Step 1: Reductions.** Note that Grothendieck's inequality becomes trivial if we allow the value of K depend on the matrix $A = (a_{ij})$. (For example, $K = \sum_{i,j} |a_{ij}|$ would work – check!) Let us choose $K = K(A)$ to be the *smallest* number that makes the conclusion (3.13) valid for a given matrix A and any Hilbert space H and any vectors $u_i, v_j \in H$. Our goal is to show that K does *not* depend on the matrix A or the dimensions m and n .

Without loss of generality,⁴ we may do this for a specific Hilbert space H , namely for $\mathbb{R}^N$ equipped with the Euclidean norm $\|\cdot\|_2$. Let us fix vectors $u_i, v_j \in \mathbb{R}^N$ which realize the smallest K , that is

$$\sum_{i,j} a_{ij} \langle u_i, v_j \rangle = K, \quad \|u_i\|_2 = \|v_j\|_2 = 1.$$

Step 2: Introducing randomness. The main idea of the proof is to realize the vectors u_i, v_j via Gaussian random variables

$$U_i := \langle g, u_i \rangle \quad \text{and} \quad V_j := \langle g, v_j \rangle, \quad \text{where } g \sim N(0, I_N).$$

As we noted in Exercise 3.3.5, U_i and V_j are standard normal random variables whose correlations follow exactly the inner products of the vectors u_i and v_j :

$$\mathbb{E} U_i V_j = \langle u_i, v_j \rangle.$$

Thus

$$K = \sum_{i,j} a_{ij} \langle u_i, v_j \rangle = \mathbb{E} \sum_{i,j} a_{ij} U_i V_j. \quad (3.14)$$

Assume for a moment that the random variables U_i and V_j were bounded almost surely by some constant – say, by R . Then the assumption (3.12) of Grothendieck's inequality (after rescaling) would yield $\left| \sum_{i,j} a_{ij} U_i V_j \right| \leq R^2$ almost surely, and (3.14) would then give $K \leq R^2$.

Step 3: Truncation. Of course, this reasoning is flawed: the random variables $U_i, V_j \sim N(0, 1)$ are not bounded almost surely. To fix this argument, we can utilize a useful *truncation* trick. Let us fix some level $R \geq 1$ and decompose the random variables as follows:

$$U_i = U_i^- + U_i^+ \quad \text{where} \quad U_i^- = U_i \mathbf{1}_{\{|U_i| \leq R\}} \quad \text{and} \quad U_i^+ = U_i \mathbf{1}_{\{|U_i| > R\}}.$$

We similarly decompose $V_j = V_j^- + V_j^+$. Now U_i^- and V_j^- are bounded by R almost surely as we desired. The remainder terms U_i^+ and V_j^+ are small in the L^2 norm: indeed, the bound in Exercise 2.1.4 gives

$$\|U_i^+\|_{L^2}^2 \leq 2 \left(R + \frac{1}{R} \right) \frac{1}{\sqrt{2\pi}} e^{-R^2/2} < \frac{4}{R^2}, \quad (3.15)$$

⁴ To see this, we can first trivially replace H with the subspace of H spanned by the vectors u_i and v_j (and with the norm inherited from H). This subspace has dimension at most $N := m + n$. Next, we recall the basic fact that all N -dimensional Hilbert spaces are isometric with each other, and in particular they are isometric to $\mathbb{R}^N$ with the norm $\|\cdot\|_2$. The isometry can be constructed by identifying orthogonal bases of those spaces.

and similarly for V_j^+ .

Step 4: Breaking up the sum. The sum in (3.14) becomes

$$K = \mathbb{E} \sum_{i,j} a_{ij} (U_i^- + U_i^+) (V_j^- + V_j^+).$$

When we expand the product in each term we obtain four sums, which we proceed to bound individually. The first sum,

$$S_1 := \mathbb{E} \sum_{i,j} a_{ij} U_i^- V_j^-,$$

is the best of all. By construction, the random variables U_i^- and V_j^- are bounded almost surely by R . Thus, just like we explained above, we can use the assumption (3.12) of Grothendieck's inequality to get $S_1 \leq R^2$.

We are not able to use the same reasoning for the second sum,

$$S_2 := \mathbb{E} \sum_{i,j} a_{ij} U_i^+ V_j^-,$$

since the random variable U_i^+ is unbounded. Instead, we will view the random variables U_i^+ and V_j^- as elements of the Hilbert space L^2 with the inner product $\langle X, Y \rangle_{L^2} = \mathbb{E} XY$. The second sum becomes

$$S_2 = \sum_{i,j} a_{ij} \langle U_i^+, V_j^- \rangle_{L^2}. \quad (3.16)$$

Recall from (3.15) that $\|U_i^+\|_{L^2} < 2/R$ and $\|V_j^-\|_{L^2} \leq \|V_j\|_{L^2} = 1$ by construction. Then, applying the conclusion (3.13) of Grothendieck's inequality for the Hilbert space $H = L^2$, we find that⁵

$$S_2 \leq K \cdot \frac{2}{R}.$$

The third and fourth sums, $S_3 := \mathbb{E} \sum_{i,j} a_{ij} U_i^- V_j^+$ and $S_4 := \mathbb{E} \sum_{i,j} a_{ij} U_i^+ V_j^+$, can be both bounded just like S_2 . (Check!)

Step 5: Putting everything together. Putting the four sums together, we conclude from (3.14) that

$$K \leq R^2 + \frac{6K}{R}.$$

Choosing $R = 12$ (for example) and solve the resulting inequality, we obtain $K \leq 288$. The theorem is proved. $\square$

Exercise 3.5.3 (Symmetric matrices, $x_i = y_i$).  Deduce the following version of Grothendieck's inequality for symmetric $n \times n$ matrices $A = (a_{ij})$ with

⁵ It might seem weird that we are able to apply the inequality that we are trying to prove. Remember, however, that we chose K in the beginning of the proof as the best number that makes Grothendieck's inequality valid. This is the K we are using here.

real entries. Suppose that A is either positive semidefinite or has zero diagonal. Assume that, for any numbers $x_i \in \{-1, 1\}$, we have

$$\left| \sum_{i,j} a_{ij} x_i x_j \right| \leq 1.$$

Then, for any Hilbert space H and any vectors $u_i, v_j \in H$ satisfying $\|u_i\| = \|v_j\| = 1$, we have

$$\left| \sum_{i,j} a_{ij} \langle u_i, v_j \rangle \right| \leq 2K, \quad (3.17)$$

where K is the absolute constant from Grothendieck's inequality.

Hint: Check and use the polarization identity $\langle Ax, y \rangle = \langle Au, u \rangle - \langle Av, v \rangle$ where $u = (x + y)/2$ and $v = (x - y)/2$.

3.5.1 Semidefinite programming

One application area where Grothendieck's inequality can be particularly helpful is the analysis of certain computationally hard problems. A powerful approach to such problems is to try and *relax* them to computationally simpler and more tractable problems. This is often done using semidefinite programming, with Grothendieck's inequality guaranteeing the quality of such relaxations.

Definition 3.5.4. A *semidefinite program* is an optimization problem of the following type:

$$\text{maximize } \langle A, X \rangle : \quad X \succeq 0, \quad \langle B_i, X \rangle = b_i \text{ for } i = 1, \dots, m. \quad (3.18)$$

Here A and B_i are given $n \times n$ matrices and b_i are given real numbers. The running “variable” X is an $n \times n$ symmetric positive semidefinite matrix, indicated by the notation $X \succeq 0$. The inner product

$$\langle A, X \rangle = \text{tr}(A^T X) = \sum_{i,j=1}^n A_{ij} X_{ij} \quad (3.19)$$

is the canonical inner product on the space of $n \times n$ matrices.

Note in passing that if we *minimize* instead of maximize in (3.18), we still get a semidefinite program. (To see this, replace A with $-A$.)

Every semidefinite program is a *convex program*, which maximizes a linear function $\langle A, X \rangle$ over a convex set of matrices. Indeed, the set of positive semidefinite matrices is convex (why?), and so is its intersection with the linear subspace defined by the constraints $\langle B_i, X \rangle = b_i$.

This is good news since convex programs are generally algorithmically tractable. There is a variety of computationally efficient solvers available for general convex programs, and for semidefinite programs (3.18) in particular, for example interior point methods.

Semidefinite relaxations

Semidefinite programs can be designed to provide computationally efficient relaxations of computationally hard problems, such as this one:

$$\text{maximize } \sum_{i,j=1}^n A_{ij} x_i x_j : \quad x_i = \pm 1 \text{ for } i = 1, \dots, n \quad (3.20)$$

where A is a given $n \times n$ symmetric matrix. This is an *integer optimization problem*. The feasible set consists of 2^n vectors $x = (x_i) \in \{-1, 1\}^n$, so finding the maximum by exhaustive search would take exponential time. Is there a smarter way to solve the problem? This is not likely: the problem (3.20) is known to be computationally hard in general (NP-hard).

Nonetheless, we can “relax” the problem (3.20) to a semidefinite program that can compute the maximum *approximately*, up to a constant factor. To formulate such a relaxation, let us replace in (3.20) the numbers $x_i = \pm 1$ by their higher-dimensional analogs – unit vectors X_i in $\mathbb{R}^n$. Thus we consider the following optimization problem:

$$\text{maximize } \sum_{i,j=1}^n A_{ij} \langle X_i, X_j \rangle : \quad \|X_i\|_2 = 1 \text{ for } i = 1, \dots, n. \quad (3.21)$$

Exercise 3.5.5. ☞☞ Show that the optimization (3.21) is equivalent to the following semidefinite program:

$$\text{maximize } \langle A, X \rangle : \quad X \succeq 0, \quad X_{ii} = 1 \text{ for } i = 1, \dots, n. \quad (3.22)$$

Hint: Consider the *Gram matrix* of the vectors X_i , which is the $n \times n$ matrix with entries $\langle X_i, X_j \rangle$. Do not forget to describe how to translate a solution of (3.22) into a solution of (3.21).

The guarantee of relaxation

We now see how Grothendieck’s inequality guarantees the accuracy of semidefinite relaxations: the semidefinite program (3.21) approximates the maximum value in the integer optimization problem (3.20) up to an absolute constant factor.

Theorem 3.5.6. *Consider an $n \times n$ symmetric, positive semidefinite matrix A . Let $\text{INT}(A)$ denote the maximum in the integer optimization problem (3.20) and $\text{SDP}(A)$ denote the maximum in the semidefinite problem (3.21). Then*

$$\text{INT}(A) \leq \text{SDP}(A) \leq 2K \cdot \text{INT}(A)$$

where $K \leq 1.783$ is the constant in Grothendieck’s inequality.

Proof The first bound follows with $X_i = (x_i, 0, 0, \dots, 0)^\top$. The second bound follows from Grothendieck’s inequality for symmetric matrices in Exercise 3.5.3. (Argue that one can drop absolute values.) $\square$

Although Theorem 3.5.6 allows us to approximate the maximum value in (3.20), it is not obvious how to compute x_i ’s that attain this approximate value. Can we translate the vectors (X_i) that give a solution of the semidefinite program

(3.21) into labels $x_i = \pm 1$ that approximately solve (3.20)? In the next section, we illustrate this on the example of a remarkable NP-hard problem on graphs – the maximum cut problem.

Exercise 3.5.7. ☛☛☛ Let A be an $m \times n$ matrix. Consider the optimization problem

$$\text{maximize } \sum_{i,j} A_{ij} \langle X_i, Y_j \rangle : \quad \|X_i\|_2 = \|Y_j\|_2 = 1 \text{ for all } i, j$$

over $X_i, Y_j \in \mathbb{R}^k$ and $k \in \mathbb{N}$. Formulate this problem as a semidefinite program.

Hint: First, express the objective function as $\frac{1}{2} \text{tr}(\tilde{A}ZZ^T)$, where $\tilde{A} = \begin{bmatrix} 0 & A \\ A^T & 0 \end{bmatrix}$, $Z = \begin{bmatrix} X \\ Y \end{bmatrix}$ and X and Y are the matrices with rows X_i^T and Y_j^T , respectively. Then express the set of matrices of the type ZZ^T with unit rows as the set of symmetric positive semidefinite matrices whose diagonal entries equal 1.

3.6 Application: Maximum cut for graphs

We now illustrate the utility of semidefinite relaxations for the problem of finding the *maximum cut* of a graph, which is one of the well known NP-hard problems discussed in the computer science literature.

3.6.1 Graphs and cuts

An undirected *graph* $G = (V, E)$ is defined as a set V of vertices together with a set E of edges; each edge is an unordered pair of vertices. Here we consider finite, *simple* graphs – those with finitely many vertices and with no loops or multiple edges.

Definition 3.6.1 (Maximum cut). Suppose we partition the set of vertices of a graph G into two disjoint sets. The *cut* is the number of edges crossing between these two sets. The maximum cut of G , denoted $\text{MAX-CUT}(G)$, is obtained by maximizing the cut over all partitions of vertices; see Figure 3.10 for illustration.

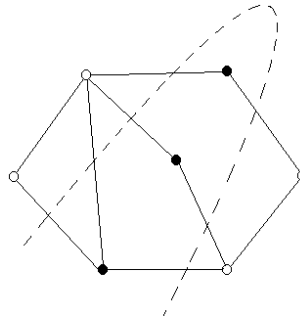


Figure 3.10 The dashed line illustrates the maximum cut of this graph, obtained by partitioning the vertices into the black and white ones. Here $\text{MAX-CUT}(G) = 7$.

Computing the maximum cut of a given graph is known to be a computationally hard problem (NP-hard).

3.6.2 A simple 0.5-approximation algorithm

We try to relax the maximum cut problem to a semidefinite program following the method we introduced in Section 3.5.1. To do this, we need to translate the problem into the language of linear algebra.

Definition 3.6.2 (Adjacency matrix). The *adjacency matrix* A of a graph G on n vertices is a symmetric $n \times n$ matrix whose entries are defined as $A_{ij} = 1$ if the vertices i and j are connected by an edge and $A_{ij} = 0$ otherwise.

Let us label the vertices of G by the integers $1, \dots, n$. A partition of the vertices into two sets can be described using a vector of labels

$$x = (x_i) \in \{-1, 1\}^n,$$

the sign of x_i indicating which subset the vertex i belongs to. For example, the three black vertices in Figure 3.10 may have labels $x_i = 1$, and the four white vertices labels $x_i = -1$. The cut of G corresponding to the partition given by x is simply the number of edges between the vertices with labels of opposite signs, i.e.

$$\text{CUT}(G, x) = \frac{1}{2} \sum_{i,j: x_i x_j = -1} A_{ij} = \frac{1}{4} \sum_{i,j=1}^n A_{ij} (1 - x_i x_j). \quad (3.23)$$

(The factor $\frac{1}{2}$ prevents double counting of edges (i, j) and (j, i) .) The maximum cut is then obtained by maximizing $\text{CUT}(G, x)$ over all x , that is

$$\text{MAX-CUT}(G) = \frac{1}{4} \max \left\{ \sum_{i,j=1}^n A_{ij} (1 - x_i x_j) : x_i = \pm 1 \text{ for all } i \right\}. \quad (3.24)$$

Let us start with a simple 0.5-approximation algorithm for maximum cut – one which finds a cut with at least *half* of the edges of G .

Proposition 3.6.3 (0.5-approximation algorithm for maximum cut). *Partition the vertices of G into two sets at random, uniformly over all 2^n partitions. Then the expectation of the resulting cut equals*

$$0.5|E| \geq 0.5 \text{MAX-CUT}(G),$$

where $|E|$ denotes the total number of edges of G .

Proof The random cut is generated by a symmetric Bernoulli random vector $x \sim \text{Unif}(\{-1, 1\}^n)$, which has independent symmetric Bernoulli coordinates. Then, in (3.23) we have $\mathbb{E} x_i x_j = 0$ for $i \neq j$ and $A_{ij} = 0$ for $i = j$ (since the graph has no loops). Thus, using linearity of expectation, we get

$$\mathbb{E} \text{CUT}(G, x) = \frac{1}{4} \sum_{i,j=1}^n A_{ij} = \frac{1}{2} |E|.$$

This completes the proof. $\square$

Exercise 3.6.4. ☕☕ For any $\varepsilon > 0$, give an $(0.5 - \varepsilon)$ -approximation algorithm for maximum cut, which is always *guaranteed* to give a suitable cut, but may have a random running time. Give a bound on the expected running time.

Hint: Consider cutting G repeatedly. Give a bound on the expected number of experiments.

3.6.3 Semidefinite relaxation

Now we will do much better and give a 0.878-approximation algorithm, which is due to Goemans and Williamson. It is based on a semidefinite relaxation of the NP-hard problem (3.24). It should be easy to guess what such relaxation could be: recalling (3.21), it is natural to consider the semidefinite problem

$$\text{SDP}(G) := \frac{1}{4} \max \left\{ \sum_{i,j=1}^n A_{ij}(1 - \langle X_i, X_j \rangle) : X_i \in \mathbb{R}^n, \|X_i\|_2 = 1 \text{ for all } i \right\}. \quad (3.25)$$

(Again – why is this a semidefinite program?)

As we will see, not only the value $\text{SDP}(G)$ approximates $\text{MAX-CUT}(G)$ to within the 0.878 factor, but we can obtain an actual partition of G (i.e., the labels x_i) which attains this value. To do this, we describe how to translate a solution (X_i) of (3.25) into labels $x_i = \pm 1$.

This can be done by the following *randomized rounding* step. Choose a random hyperplane in $\mathbb{R}^n$ passing through the origin. It cuts the set of vectors X_i into two parts; let us assign labels $x_i = 1$ to one part and $x_i = -1$ to the other part. Equivalently, we may choose a standard normal random vector

$$g \sim N(0, I_n)$$

and define

$$x_i := \text{sign} \langle X_i, g \rangle, \quad i = 1, \dots, n. \quad (3.26)$$

See Figure 3.11 for an illustration.⁶

Theorem 3.6.5 (0.878-approximation algorithm for maximum cut). *Let G be a graph with adjacency matrix A . Let $x = (x_i)$ be the result of a randomized rounding of the solution (X_i) of the semidefinite program (3.25). Then*

$$\mathbb{E} \text{CUT}(G, x) \geq 0.878 \text{SDP}(G) \geq 0.878 \text{MAX-CUT}(G).$$

The proof of this theorem will be based on the following elementary identity. We can think of it as a more advanced version of the identity (3.6), which we used in the proof of Grothendieck's inequality, Theorem 3.5.1.

⁶ In the rounding step, instead of the normal distribution we could use any other rotation invariant distribution in $\mathbb{R}^n$, for example the uniform distribution on the sphere S^{n-1} .

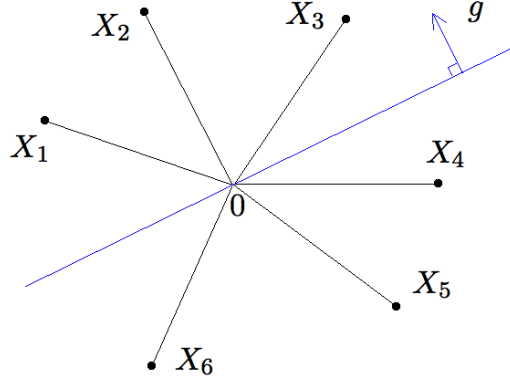


Figure 3.11 Randomized rounding of vectors $X_i \in \mathbb{R}^n$ into labels $x_i = \pm 1$. For this configuration of points X_i and a random hyperplane with normal vector g , we assign $x_1 = x_2 = x_3 = 1$ and $x_4 = x_5 = x_6 = -1$.

Lemma 3.6.6 (Grothendieck's identity). *Consider a random vector $g \sim N(0, I_n)$. Then, for any fixed vectors $u, v \in S^{n-1}$, we have*

$$\mathbb{E} \operatorname{sign} \langle g, u \rangle \operatorname{sign} \langle g, v \rangle = \frac{2}{\pi} \arcsin \langle u, v \rangle.$$

Exercise 3.6.7. ☕☕ Prove Grothendieck's identity.

Hint: It will quickly follow once you show that the probability that $\langle g, u \rangle$ and $\langle g, v \rangle$ have opposite signs equals α/π , where $\alpha \in [0, \pi]$ is the angle between the vectors u and v . To check this, use rotation invariance to reduce the problem to $\mathbb{R}^2$. Once on the plane, rotation invariance will give the result.

A weak point of Grothendieck's identity is the non-linear function $\arcsin$, which would be hard to work with. Let us replace it with a linear function using the numeric inequality

$$1 - \frac{2}{\pi} \arcsin t = \frac{2}{\pi} \arccos t \geq 0.878(1 - t), \quad t \in [-1, 1], \quad (3.27)$$

which can be easily verified using software; see Figure 3.12.

Proof of Theorem 3.6.5 By (3.23) and linearity of expectation, we have

$$\mathbb{E} \operatorname{CUT}(G, x) = \frac{1}{4} \sum_{i,j=1}^n A_{ij} (1 - \mathbb{E} x_i x_j).$$

The definition of labels x_i in the rounding step (3.26) gives

$$\begin{aligned} 1 - \mathbb{E} x_i x_j &= 1 - \mathbb{E} \operatorname{sign} \langle X_i, g \rangle \operatorname{sign} \langle X_j, g \rangle \\ &= 1 - \frac{2}{\pi} \arcsin \langle X_i, X_j \rangle \quad (\text{by Grothendieck's identity, Lemma 3.6.6}) \\ &\geq 0.878(1 - \langle X_i, X_j \rangle) \quad (\text{by (3.27)}). \end{aligned}$$

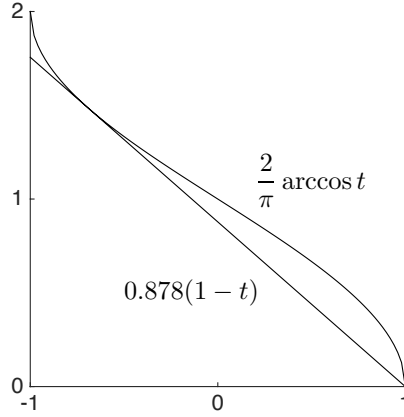


Figure 3.12 The inequality $\frac{2}{\pi} \arccos t \geq 0.878(1-t)$ holds for all $t \in [-1, 1]$.

Therefore

$$\mathbb{E} \text{CUT}(G, x) \geq 0.878 \cdot \frac{1}{4} \sum_{i,j=1}^n A_{ij}(1 - \langle X_i, X_j \rangle) = 0.878 \text{SDP}(G).$$

This proves the first inequality in the theorem. The second inequality is trivial since $\text{SDP}(G) \geq \text{MAX-CUT}(G)$. (Why?) $\square$

3.7 Kernel trick, and tightening of Grothendieck's inequality

Our proof of Grothendieck's inequality given in Section 3.5 yields a very loose bound on the absolute constant K . We now give an alternative proof that gives (almost) the best known constant $K \leq 1.783$.

Our new argument will be based on Grothendieck's identity (Lemma (3.6.6)). The main challenge in using this identity arises from the non-linearity of the function $\arcsin(x)$. Indeed, suppose there were no such nonlinearity, and we hypothetically had $\mathbb{E} \text{sign} \langle g, u \rangle \text{sign} \langle g, v \rangle = \frac{2}{\pi} \langle u, v \rangle$. Then Grothendieck's inequality would easily follow:

$$\frac{2}{\pi} \sum_{i,j} a_{ij} \langle u_i, v_j \rangle = \sum_{i,j} a_{ij} \mathbb{E} \text{sign} \langle g, u_i \rangle \text{sign} \langle g, v_j \rangle \leq 1,$$

where in the last step we swapped the sum and expectation and used the assumption of Grothendieck's inequality with $x_i = \text{sign} \langle g, u_i \rangle$ and $y_j = \text{sign} \langle g, v_j \rangle$. This would give Grothendieck's inequality with $K \leq \pi/2 \approx 1.57$.

This argument is of course wrong. To address the non-linear form $\frac{2}{\pi} \arcsin \langle u, v \rangle$ that appears in Grothendieck's identity, we use the following remarkably powerful trick: represent $\frac{2}{\pi} \arcsin \langle u, v \rangle$ as the (linear) inner product $\langle u', v' \rangle$ of some other

vectors u', v' in some Hilbert space H . In the literature on machine learning, this method is called the *kernel trick*.

We will explicitly construct the non-linear transformations $u' = \Phi(u)$, $v' = \Psi(v)$ that will do the job. Our construction is convenient to describe in the language of *tensors*, which are a higher dimensional generalization of the notion of matrices.

Definition 3.7.1 (Tensors). A tensor can be described as a multidimensional array. Thus, a k -th order tensor $(a_{i_1 \dots i_k})$ is a k -dimensional array of real numbers $a_{i_1 \dots i_k}$. The canonical inner product on $\mathbb{R}^{n_1 \times \dots \times n_k}$ defines the inner product of tensors $A = (a_{i_1 \dots i_k})$ and $B = (b_{i_1 \dots i_k})$:

$$\langle A, B \rangle := \sum_{i_1, \dots, i_k} a_{i_1 \dots i_k} b_{i_1 \dots i_k}. \quad (3.28)$$

Example 3.7.2. Scalars, vectors and matrices are examples of tensors. As we noted in (3.19), for $m \times n$ matrices the inner product of tensors (3.28) specializes to

$$\langle A, B \rangle = \text{tr}(A^T B) = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}.$$

Example 3.7.3 (Rank-one tensors). Every vector $u \in \mathbb{R}^n$ defines the k -th order *tensor product* $u \otimes \dots \otimes u$, which is the tensor whose entries are the products of all k -tuples of the entries of u . In other words,

$$u \otimes \dots \otimes u = u^{\otimes k} := (u_{i_1} \dots u_{i_k}) \in \mathbb{R}^{n \times \dots \times n}.$$

In particular, for $k = 2$, the tensor product $u \otimes u$ is just the $n \times n$ matrix which is the outer product of u with itself:

$$u \otimes u = (u_i u_j)_{i,j=1}^n = uu^T.$$

One can similarly define the tensor products $u \otimes v \otimes \dots \otimes z$ for different vectors $u, v, \dots, z$.

Exercise 3.7.4. ☞ Show that for any vectors $u, v \in \mathbb{R}^n$ and $k \in \mathbb{N}$, we have

$$\langle u^{\otimes k}, v^{\otimes k} \rangle = \langle u, v \rangle^k.$$

This exercise shows a remarkable fact: we can represent non-linear forms like $\langle u, v \rangle^k$ as the usual, *linear* inner product in some other space. Formally, there exist a Hilbert space H and a transformation $\Phi : \mathbb{R}^n \rightarrow H$ such that

$$\langle \Phi(u), \Phi(v) \rangle = \langle u, v \rangle^k.$$

In this case, H is the space of k -th order tensors, and $\Phi(u) = u^{\otimes k}$.

In the next two exercises, we extend this observation to more general non-linearities.

Exercise 3.7.5. ☞☞

- (a) Show that there exist a Hilbert space H and a transformation $\Phi : \mathbb{R}^n \rightarrow H$ such that

$$\langle \Phi(u), \Phi(v) \rangle = 2 \langle u, v \rangle^2 + 5 \langle u, v \rangle^3 \quad \text{for all } u, v \in \mathbb{R}^n.$$

Hint: Consider the cartesian product $H = \mathbb{R}^{n \times n} \oplus \mathbb{R}^{n \times n \times n}$.

- (b) More generally, consider a polynomial $f : \mathbb{R} \rightarrow \mathbb{R}$ with non-negative coefficients, and construct H and Φ such that

$$\langle \Phi(u), \Phi(v) \rangle = f(\langle u, v \rangle) \quad \text{for all } u, v \in \mathbb{R}^n.$$

- (c) Show the same for any *real analytic function* $f : \mathbb{R} \rightarrow \mathbb{R}$ with non-negative coefficients, i.e. for any function that can be represented as a convergent series

$$f(x) = \sum_{k=0}^{\infty} a_k x^k, \quad x \in \mathbb{R}, \quad (3.29)$$

and such that $a_k \geq 0$ for all k .

Exercise 3.7.6.  Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be any real analytic function (with possibly negative coefficients in (3.29)). Show that there exist a Hilbert space H and transformations $\Phi, \Psi : \mathbb{R}^n \rightarrow H$ such that

$$\langle \Phi(u), \Psi(v) \rangle = f(\langle u, v \rangle) \quad \text{for all } u, v \in \mathbb{R}^n.$$

Moreover, check that

$$\|\Phi(u)\|^2 = \|\Psi(u)\|^2 = \sum_{k=0}^{\infty} |a_k| \|u\|_2^{2k}.$$

Hint: Construct Φ as in Exercise 3.7.5 with Φ , but include the signs of a_k in the definition of Ψ .

Let us specialize the kernel trick to the non-linearity $\frac{2}{\pi} \arcsin \langle u, v \rangle$ that appears in Grothendieck's identity.

Lemma 3.7.7. *There exists a Hilbert space H and transformations⁷ $\Phi, \Psi : S^{n-1} \rightarrow S(H)$ such that*

$$\frac{2}{\pi} \arcsin \langle \Phi(u), \Psi(v) \rangle = \beta \langle u, v \rangle \quad \text{for all } u, v \in S^{n-1}, \quad (3.30)$$

where $\beta = \frac{2}{\pi} \ln(1 + \sqrt{2})$.

Proof Rewrite the desired identity (3.30) as

$$\langle \Phi(u), \Psi(v) \rangle = \sin \left(\frac{\beta\pi}{2} \langle u, v \rangle \right). \quad (3.31)$$

The result of Exercise 3.7.6 gives us the Hilbert space H and the maps $\Phi, \Psi :$

⁷ Here $S(H)$ denotes the unit sphere of the Hilbert space H .

$\mathbb{R}^n \rightarrow H$ that satisfy (3.31). It only remains to determine the value of β for which Φ and Ψ map unit vectors to unit vectors. To do this, we recall the Taylor series

$$\sin t = t - \frac{t^3}{3!} + \frac{t^5}{5!} - \cdots \quad \text{and} \quad \sinh t = t + \frac{t^3}{3!} + \frac{t^5}{5!} + \cdots$$

Exercise 3.7.6 then guarantees that for every $u \in S^{n-1}$, we have

$$\|\Phi(u)\|^2 = \|\Psi(u)\|^2 = \sinh\left(\frac{\beta\pi}{2}\right).$$

This quantity equals 1 if we set

$$\beta := \frac{2}{\pi} \operatorname{arcsinh}(1) = \frac{2}{\pi} \ln(1 + \sqrt{2}).$$

The lemma is proved. $\square$

Now we are ready to prove Grothendieck's inequality (Theorem 3.5.1) with constant

$$K \leq \frac{1}{\beta} = \frac{\pi}{2 \ln(1 + \sqrt{2})} \approx 1.783.$$

Proof of Theorem 3.5.1 We can assume without loss of generality that $u_i, v_j \in S^{N-1}$ (this is the same reduction as we did in the proof in Section 3.5). Lemma 3.7.7 gives us unit vectors $u'_i = \Phi(u_i)$ and $v'_j = \Psi(v_j)$ in some Hilbert space H , which satisfy

$$\frac{2}{\pi} \arcsin \langle u'_i, v'_j \rangle = \beta \langle u_i, v_j \rangle \quad \text{for all } i, j.$$

We can again assume without loss of generality that $H = \mathbb{R}^M$ for some M . (Why?) Then

$$\begin{aligned} \beta \sum_{i,j} a_{ij} \langle u_i, v_j \rangle &= \sum_{i,j} a_{ij} \cdot \frac{2}{\pi} \arcsin \langle u'_i, v'_j \rangle \\ &= \sum_{i,j} a_{ij} \mathbb{E} \operatorname{sign} \langle g, u'_i \rangle \operatorname{sign} \langle g, v'_j \rangle \quad (\text{by Lemma 3.6.6}), \\ &\leq 1, \end{aligned}$$

where in the last step we swapped the sum and expectation and used the assumption of Grothendieck's inequality with $x_i = \operatorname{sign} \langle g, u'_i \rangle$ and $y_j = \operatorname{sign} \langle g, v'_j \rangle$. This yields the conclusion of Grothendieck's inequality for $K \leq 1/\beta$. $\square$

3.7.1 Kernels and feature maps

Since the kernel trick was so successful in the proof of Grothendieck's inequality, we may ask – what other non-linearities can be handled with the kernel trick? Let

$$K : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$$

be a function of two variables on a set $\mathcal{X}$. Under what conditions on K can we find a Hilbert space H and a transformation

$$\Phi : \mathcal{X} \rightarrow H$$

so that

$$\langle \Phi(u), \Phi(v) \rangle = K(u, v) \quad \text{for all } u, v \in \mathcal{X} \quad (3.32)$$

The answer to this question is provided by Mercer's and, more precisely, Moore-Aronszajn's theorems. The necessary and sufficient condition is that K be a *positive semidefinite kernel*, which means that for any finite collection of points $u_1, \dots, u_N \in \mathcal{X}$, the matrix

$$\left(K(u_i, u_j) \right)_{i,j=1}^N$$

is symmetric and positive semidefinite. The map Φ is called a *feature map*, and the Hilbert space H can be constructed from the kernel K as a (unique) *reproducing kernel Hilbert space*.

Examples of positive semidefinite kernels on $\mathbb{R}^n$ that are common in machine learning include the *Gaussian kernel* (also called the radial basis function kernel)

$$K(u, v) = \exp \left(- \frac{\|u - v\|_2^2}{2\sigma^2} \right), \quad u, v \in \mathbb{R}^n, \sigma > 0$$

and the *polynomial kernel*

$$K(u, v) = \left(\langle u, v \rangle + r \right)^k, \quad u, v \in \mathbb{R}^n, r > 0, k \in \mathbb{N}.$$

The kernel trick (3.32), which represents a general kernel $K(u, v)$ as an inner product, is very popular in *machine learning*. It allows one to handle non-linear models (determined by kernels K) by using methods developed for linear models. In contrast to what we did in this section, in machine learning applications the explicit description of the Hilbert space H and the feature map $\Phi : \mathcal{X} \rightarrow H$ is typically not needed. Indeed, to compute the inner product $\langle \Phi(u), \Phi(v) \rangle$ in H , one does not need to know Φ : the identity (3.32) allows one to compute $K(u, v)$ instead.

3.8 Notes

Theorem 3.1.1 about the concentration of the norm of random vectors is known but difficult to locate in the existing literature. We will later prove a more general result, Theorem 6.3.2, which is valid for anisotropic random vectors. It is unknown if the quadratic dependence on K in Theorem 3.1.1 is optimal. One may also wonder about concentration of the norm $\|X\|_2$ of random vectors X whose coordinates are not necessarily independent. In particular, for a random vector X that is uniformly distributed in a convex set K , concentration of the norm is one of the central problems in geometric functional analysis; see [93, Section 2] and [36, Chapter 12].

Exercise 3.3.4 mentions Cramér-Wold’s theorem. It is a straightforward consequence of the uniqueness theorem for characteristic functions, see [23, Section 29].

The concept of frames introduced in Section 3.3.4 is an important extension of the notion of orthogonal bases. One can read more about frames and their applications in signal processing and data compression e.g. in [51, 121].

Sections 3.3.5 and 3.4.4 discuss random vectors uniformly distributed in convex sets. The books [11, 36] study this topic in detail, and the surveys [185, 218] discuss algorithmic aspects of computing the volume of convex sets in high dimensions.

Our discussion of sub-gaussian random vectors in Section 3.4. mostly follows [222]. An alternative geometric proof of Theorem 3.4.6 can be found in [13, Lemma 2.2].

Grothendieck’s inequality (Theorem 3.5.1) was originally proved by A. Grothendieck in 1953 [90] with bound on the constant $K \leq \sinh(\pi/2) \approx 2.30$; a version of this original argument is presented [133, Section 2]. There is a number of alternative proofs of Grothendieck’s inequality with better and worse bounds on K ; see [35] for the history. The surveys [115, 168] discuss ramifications and applications of Grothendieck’s inequality in various areas of mathematics and computer science. Our first proof of Grothendieck’s inequality, the one given in Section 3.5, is similar to the one in [5, Section 8.1]; it was kindly brought to author’s attention by Mark Rudelson. Our second proof, the one from Section 3.7, is due to J.-L. Krivine [122]; versions of this argument can be found e.g. in [7] and [126]. The bound on the constant $K \leq \frac{\pi}{2 \ln(1+\sqrt{2})} \approx 1.783$ that follows from Krivine’s argument is currently the best known *explicit* bound on K . It has been proved, however, that the best possible bound must be strictly smaller than Krivine’s bound, but no explicit number is known [35].

A part of this chapter is about semidefinite relaxations of hard optimization problems. For an introduction to the area of convex optimization, including semidefinite programming, we refer to the books [34, 39, 126, 29]. For the use of Grothendieck’s inequality in analyzing semidefinite relaxations, see [115, 7]. Our presentation of the maximum cut problem in Section 3.6 follows [39, Section 6.6] and [126, Chapter 7]. The semidefinite approach to maximum cut, which we discussed in Section 3.6.3, was pioneered in 1995 by M. Goemans and D. Williamson [83]. The approximation ratio $\frac{2}{\pi} \min_{0 \leq \theta \leq \pi} \frac{\theta}{1 - \cos(\theta)} \approx 0.878$ guaranteed by the Goemans-Williamson algorithm remains the best known constant for the max-cut problem. If the Unique Games Conjecture is true, this ratio can not be improved, i.e. any better approximation would be NP-hard to compute [114].

In Section 3.7 we give Krivine’s proof of Grothendieck’s inequality [122]. We also briefly discuss kernel methods there. To learn more about kernels, reproducing kernel Hilbert spaces and their applications in machine learning, see e.g. the survey [102].